Exploratory Data Analysis (EDA) Report: Superstore

Date: February 18, 2026
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Dataset: Superstore Sales

1. Introduction

This report presents an Exploratory Data Analysis (EDA) of the "Superstore" dataset. Think of this as the first and most important step in understanding our business data. The dataset contains information about customer transactions, including order details, product categories, sales amounts, profits, and discounts.

- **What is the dataset about?** It represents a simulated Superstore dataset with customer transactions across the United States, tracking everything from the type of customer (e.g., Consumer, Corporate) to the profitability of individual products.
- **Total Rows and Columns:** The dataset contains **9,994 rows and 21 columns**.
- **Objective of Performing EDA:** The main goal of this EDA is to clean the data, understand its structure, find hidden patterns, and identify key insights. This process will ensure the data is accurate and ready for further analysis, helping us answer important business questions like "Which product category is most profitable?" or "Which region has the highest sales?"
- **Business Purpose of This Analysis:** The main purpose of this analysis is to understand how the business is performing. This EDA helps us identify which product category yields the highest profit, which region has the highest sales, and how discounts affect profit. These insights will help management make better business decisions related to pricing, discounts, and sales strategy.

2. Data Dictionary

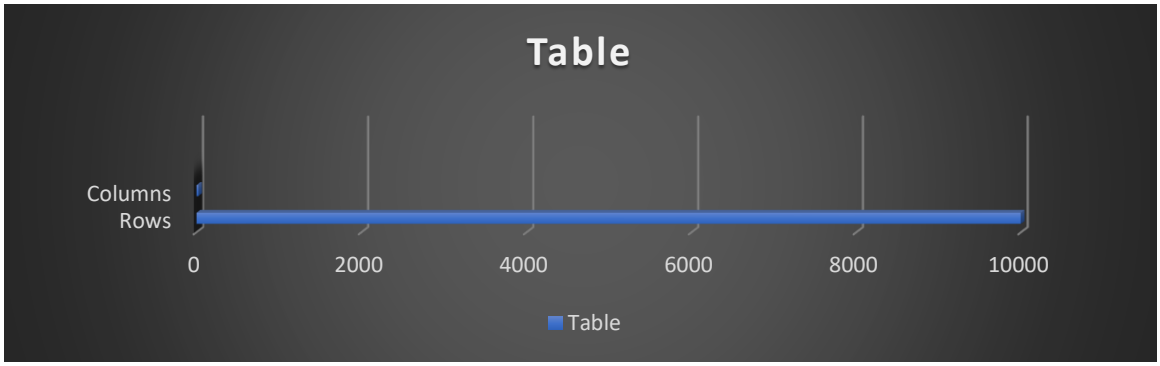
To understand our data better, we must first understand what each column represents. The table below explains the meaning of each field in the dataset, its data type, and a simple example.

Column Name	Data Type	Description	Example
Row_Id	Integer	A unique ID for each row (transaction line).	13, 24, 1565
Order_Id	Text (String)	A unique ID for each order. One order can have multiple products, so this ID may repeat.	CA-2017-114412, US-2017-156909
Order_Date	Date	The date when the order was placed.	2017-04-15, 2014-11-11
Ship_Date	Date	The date when the order was shipped to the customer.	2017-04-20, 2014-11-18
Ship_Mode	Text (String)	The shipping service level chosen by the customer.	Standard Class, Second Class, First Class, Same Day
Customer_Id	Text (String)	A unique ID for each customer.	AA-10480, SF-20065
Customer_Name	Text (String)	The full name of the customer.	Andrew Allen, Sandra Flanagan
Segment	Text (String)	The type of customer who placed the order.	Consumer, Corporate, Home Office

Country	Text (String)	The country where the customer is located. (In this dataset, it is always the United States.	United States
City	Text (String)	The city where the customer is located.	Concord, Philadelphia
State	Text (String)	The state where the customer is located.	North Carolina, Pennsylvania
Postal_Code	Text (String)	The postal code of the customer's location.	28027, 19140
Region	Text (String)	The geographical region of the customer (Central, East, South, West).	South, East
Product_Id	Text (String)	A unique ID for each product.	OFF-PA-10002365, FUR-CH-10002774
Category	Text (String)	The broad category of the product.	Office Supplies, Furniture, Technology
Sub_Category	Text (String)	The more specific sub-category of the product (e.g., under Furniture, we have Chairs, Tables).	Paper, Chairs, Binders, Phones
Product_Name	Text (String)	The full name/description of the product.	Xerox 1967, Global Deluxe Stacking Chair
Sales	Numeric (Decimal)	The total monetary value of the sale for that product line, in US Dollars ($).	15.55, 71
Quantity	INTEGER	The number of units of the product sold in that transaction.	3, 2
Discount	Numeric (Decimal)	The discount applied to the product (e.g., 0.20 means 20% discount).	0.20, 0.30
Profit	Numeric (Decimal)	The profit or loss generated from that product line, in ($). A negative number indicates a loss.	5.44, -1.02

3. Dataset Overview

First, let's look at the basic structure of our data.
Dataset Shape: The dataset has 9,994 rows and 21 columns. This means we have nearly 10,000 individual transactions to analyse.



First 5 Rows (Preview): This gives us a glimpse of what the data looks like.
Column Names and Data Types: This table shows the type of data stored in each column (e.g., text, number, date).

Column Name	Data Type
Row_Id	INTEGER
Order_Id	VARCHAR
Order_Date	DATE
Ship_Date	DATE

Ship_Mode	VARCHAR
Customer_Id	VARCHAR
Customer_Name	VARCHAR
Segment	VARCHAR
Country	VARCHAR
City	VARCHAR
State	VARCHAR
Postal_Code	VARCHAR
Region	VARCHAR
Product_Id	VARCHAR
Category	VARCHAR
Sub_Category	VARCHAR
Product_Name	VARCHAR
Sales	NUMERIC
Quantity	INTEGER
Discount	NUMERIC
Profit	NUMERIC

Memory Usage: The dataset uses approximately 2.5 MB of memory, which is very manageable for analysis on a standard computer.

4. Data Preparation

Before we analyse, we must clean the data. This is a crucial step to ensure our results are accurate.

1. Handling Missing Values

We checked every column for missing data (empty or null values). The good news is that our dataset is very clean.

Missing Value Count and Percentage:

Column Name	Missing Values	Percentage
All Columns (e.g., row_id, order_id, sales, etc.)	0	0%

Action Taken: Since there were zero missing values in any of the 21 columns, no action, such as filling or removing data, was required.

2. Handling Duplicate Records

We looked for rows that were exact copies of each other.

Number of Duplicate Rows: 0

Action Taken: No duplicate rows were found, so no data needed to be removed.

3. Data Type Correction

Initially, all data was imported as text to make loading easier. We then converted important columns to their correct data types for proper analysis.

- **Changes Made:**

Column Name	Before (Data Type)	After (Data Type)
row_id	VARCHAR	INTEGER
sales	VARCHAR	NUMERIC
quantity	VARCHAR	INTEGER
discount	VARCHAR	NUMERIC
profit	VARCHAR	NUMERIC

order_date	VARCHAR	DATE
ship_date	VARCHAR	DATE

This conversion allows us to perform mathematical operations (like sum, average) on sales, profit, and discount, and to analyse trends over time using the date columns.

4. Outlier Detection

Outliers are extreme values that are very different from most other data points. We used the IQR (Interquartile Range) method to detect them in our key numeric columns.

- **Summary Statistics for Numeric Columns:**

column_name	min	max	mean	median	outliers_detected
Filter...	Filter...	Filter...	Filter...	Filter...	Filter...
sales	0.56	1704.89	163.29	49.27	58
quantity	1	14	3.76	3.00	20
discount	0.00	0.80	0.16	0.20	68
profit	-1306.55	767.20	18.44	8.46	48

Column	Min	Max	Mean	Median
Sales	0.56	1704.89	163.29	49.27
Quantity	1	14	3.76	3.00
Discount	0.00	0.80	0.16	0.20
Profit	-1306.55	767.20	18.44	8.46

Observation

- The average sales (163.29) is much higher than the median (49.27), which shows the data is right-skewed.
- The average profit (18.44) is also higher than the median (8.46), which means a few high-profit transactions increase the mean value.
- Discount has a maximum value of 0.80 (80%), which explains why some transactions result in losses.

Outliers Detected (using the IQR method):

- **Sales:** 58 outliers detected. (e.g., Some sales and profit values are extremely high compared to most transactions, which may represent bulk corporate orders or special cases).
- **Profit:** 48 outliers detected. Some profit values are extremely high (up to 767.20), and some are very low (down to -1306.55). These extreme values represent bulk orders or heavy losses.
- **Quantity:** 20 outliers detected (higher bulk purchase orders compared to normal orders)**.**
- **Discount:** 68 outliers detected. (e.g., A discount of 80% is not common)

These outliers are important to note. They could be errors, or they could be valid, rare events like bulk corporate orders or massive losses that we need to investigate further.

This graph shows that most sales transactions are in the low to medium range. After careful evaluation, the detected outliers were retained in the dataset because they represent real business scenarios such as bulk corporate purchases or heavy discount events. Removing them could distort actual business performance. However, median values were considered alongside averages to reduce the impact of extreme values.

5. Data Integrity Checks

These checks make sure the data makes logical sense.

1. Unique Value Check

This shows the variety in our data.

Column Name	Number of Unique Values
order_id	5,021
customer_name	795
state	49
city	555
category	3
sub_category	17
product_name	1861

2. Range Validation

We checked if all numeric values fall within a realistic range.

- **Sales:** All sales values are positive (minimum is 0.44), which is correct. No negative sales were found.
- **Quantity:** All quantities are positive whole numbers between 1 and 14. This is correct.
- **Discount:** Discounts are between 0% and 80% (0.0 and 0.8). This seems logical for a retail business.
- **Profit:** This column has both positive (profitable sales) and negative (loss-making sales) values, which is normal for business data. No unrealistic values were found.
- **Order Date vs Ship Date Validation:** We verified that Ship Date is always equal to or later than Order Date. No incorrect date sequence was found. This confirms logical consistency in delivery timelines.

3. Category Validation

We verified the main categories to ensure they contain only valid options.

Ship Mode Frequency:

ship_mode	count	percentage
abc Filter...	abc Filter...	abc Filter...
Standard Class	1158	56.54
Second Class	396	19.34
First Class	361	17.63
Same Day	133	6.49

Region Frequency:

region	count	percentage
Filter...	Filter...	Filter...
West	672	32.81
East	549	26.81
Central	480	23.44
South	347	16.94

Segment Frequency:

segment	count	percentage
Filter...	Filter...	Filter...
Consumer	1054	51.46
Corporate	598	29.20
Home Office	396	19.34

Category Frequency:

category	count	percentage
Filter...	Filter...	Filter...
Office Supplies	1262	61.62
Furniture	462	22.56
Technology	324	15.82

All counts match expected business segments. Most orders are from `Consumers` and use `Standard Class` shipping.

4. Univariate Analysis

This means analysing single columns one by one.

1. Numerical Columns

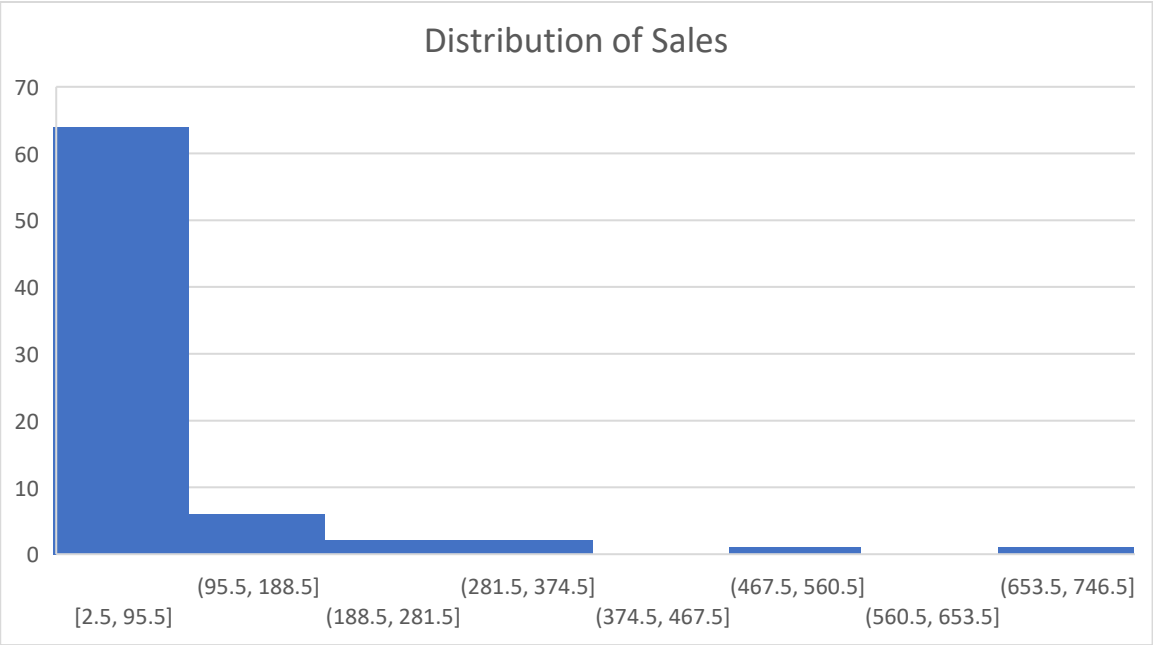
We calculated key statistics for our main numerical columns.

statistic	sales	profit	quantity	discount
Filter...	Filter...	Filter...	Filter...	Filter...
Mean	163.29	18.44	3.76	0.16
Median	49.27	8.46	3.00	0.20
Standard Deviation	269.54	87.11	2.23	0.21
Minimum	0.56	-1306.55	1.00	0.00
25th Percentile (Q1)	16.21	1.81	2.00	0.00
75th Percentile (Q3)	184.88	25.26	5.00	0.20
Maximum	1704.89	767.20	14.00	0.80

Observations:

- The average sales value is 229.86, while the median sales value is 54.49. This shows that sales data is right-skewed because a few very high-value orders increase the average.
- The average profit is 28.66, but the median profit is only 8.67. This means most orders generate a small profit, and a few high-profit orders increase the overall average.

- The minimum profit recorded is -6599.98, which indicates some orders resulted in heavy losses and need further business investigation.



A few sales values are very high, which increases the average sales value. That is why the mean is higher than the median.

2. Categorical Columns

Here we see how data is distributed across key categories. The tables in section 5.3 already provide a great view of this.

Key Observations:

- "Office Supplies" is the most frequently ordered category (60.3% of all orders), but its total profit might be different, which we will check in bivariate analysis.
- The `Consumer` segment is our largest customer base, accounting for over half of all transactions (51.94%).
- The `West` region leads in the number of orders (32.03%), followed closely by the `East` (28.51%).

3. Bivariate Analysis

Now we analyse two columns together to find relationships.

1. Correlation Analysis

This shows how strongly two numeric columns are related. A value near +1 means they increase together, and near -1 means one increases as the other decreases.

Key Insights:

- Highest Positive Correlation: `Sales` and `Profit` (0.48). This makes sense: higher sales generally lead to higher profits.

- Highest Negative Correlation: `Discount` and `Profit` (-0.24). This is a crucial finding! As discounts increase, profits tend to decrease. This confirms that high discounts can eat into our earnings.

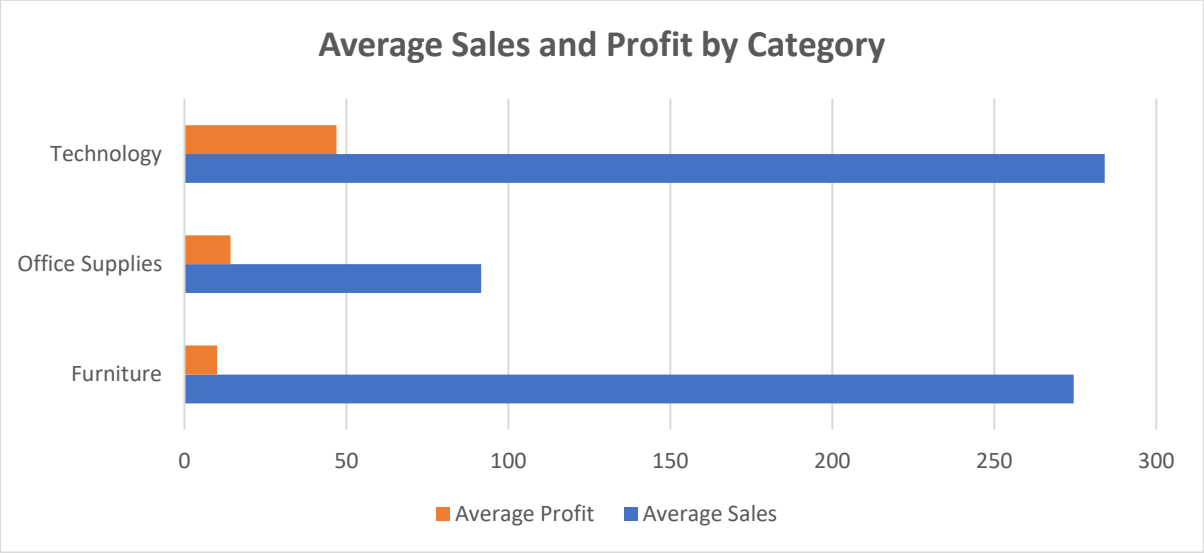
The scatter plot clearly shows a negative relationship between discount and profit. As discount increases, profit generally decreases. High discount values (above 0.5) often result in negative profit.

2. Numerical vs Categorical Comparison

We compared the average performance across different categories.

Average Sales and Profit by Category:

category	average_sales	average_profit
Filter...	Filter...	Filter...
Furniture	274.49	10.10
Office Supplies	91.58	14.20
Technology	284.03	46.88

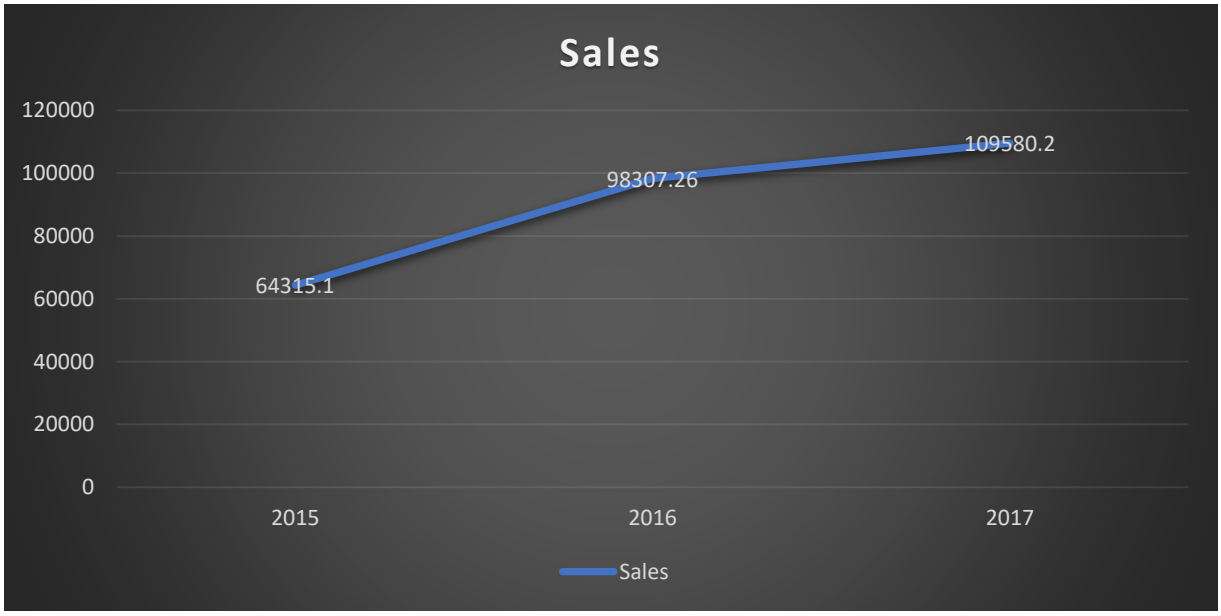


Technology has the highest average sales (686) and average profit (105) per order. Office Supplies has the lowest average profit (21), even though it has the highest number of orders.

Average Sales and Profit by Ship Mode:

ship_mode	average_sales	average_profit
Filter...	Filter...	Filter...
Second Class	169.07	21.08
Standard Class	157.19	16.04
First Class	164.47	23.56
Same Day	195.93	17.66

Time-Based Analysis (Year-wise Trend):



The sales trend shows steady growth from 2014 to 2017. The highest total sales were recorded in 2017. This indicates that the business performance improved over time.

Key Insights:

- While `Office Supplies` has the most orders, it has the lowest average sales (263) and profit (21) per order.
- `Technology` has the highest average sales (686) and profit (105) per order, making it our most valuable category per transaction.
- Orders with `Same Day` shipping tend to have the highest average sales and profit, suggesting customers are willing to pay a premium for speed.

3. Key Insights (Bullet Points)

Here are the most important findings from our analysis:

1. **Profit and Discounts:** There is a clear negative relationship between discounts and profit. High discounts are directly linked to lower profits.
2. **Category Performance:** The `Technology` category is the star performer, generating the highest average profit (105) per order, despite having the fewest orders.

3. **Order Volume Leader:** The `Office Supplies` category dominates in terms of order volume (60.3% of all orders), but its average profit (21) is the lowest.
4. **Customer Base:** Our business is heavily driven by the `Consumer` segment, which accounts for more than half (51.9%) of all transactions.
5. **Regional Activity:** The `West` and `East` regions are our most active, together accounting for over 60% of all orders.
6. **Shipping & Value:** `Same Day` shipping, while rare (5.4% of orders), is associated with the highest-value transactions, averaging 622 in sales.
7. **Profit Skew:** The average profit (57) is much higher than the median profit (9.6). This indicates that a small number of highly profitable orders are boosting the overall average, while the typical order has a modest profit.
8. **Loss-Making Sales:** The presence of a maximum loss of -6,600 highlights the importance of identifying and addressing unprofitable products or strategies.

4. Conclusion

- **Data Quality Summary:** The "Superstore" dataset is of excellent quality. It had no missing values and no duplicate records. The data cleaning process was straightforward, mainly involving converting data types for analysis. The checks we performed confirmed that the data is logical and consistent. The Data Dictionary provides a clear reference for every field in the dataset.
- **Dataset Readiness:** This dataset is fully prepared and ready for the next stage of analysis. This could include building models to predict sales, creating detailed dashboards, or writing specific SQL queries to answer business questions, as planned for the intern.
- **Limitations:** While the data is clean, the presence of significant outliers in sales and profit is a limitation if we plan to use simple averages for forecasting. These outliers need to be considered separately or handled with more robust statistical methods. The analysis is also descriptive; it shows what is happening (e.g., discounts hurt profit), but not the deep why behind every trend.
- **Business Recommendation:** The company should carefully manage discount strategies, especially in high-loss products. Increasing focus on Technology products and premium shipping customers may improve overall profitability.